

Table 1: Cross-omics Pathway Axis Validation Summary

| Pathway Axis                | Driver Gene | Evidence | N Metabolites | Match Rate | Mean log2FC | Key Metabolites                             |
|-----------------------------|-------------|----------|---------------|------------|-------------|---------------------------------------------|
| ACSL4 / Lipid signaling     | ACSL4       | Weak     | 20            | 4/20 (20%) | −0.87       | SA (18:0), PA (16:0), Ceramide, EPA (20:5)  |
| HIF1A / Hypoxia             | HIF1A       | Weak     | 2             | 0/2 (0%)   | −0.33       | Pyruvate, Succinate                         |
| IL6 / Inflammation          | IL6         | Weak     | 7             | 3/7 (43%)  | −0.21       | Cysteine, Taurine                           |
| KEAP1 / NRF2 antioxidant    | KEAP1       | Moderate | 10            | 5/10 (50%) | −0.21       | GSH (reduced), Cysteine, Taurine            |
| NAMPT / NAD+ metabolism     | NAMPT       | Weak     | 5             | 1/5 (20%)  | 0.47        | NAM                                         |
| NOX4 / Oxidative stress     | NOX4        | Weak     | 5             | 2/5 (40%)  | −0.07       | 4-HNE, GSSG, GSH (reduced)                  |
| SAT1 / Polyamine            | SAT1        | Moderate | 7             | 5/7 (71%)  | −0.34       | Ornithine, Spermidine, Putrescine, Spermine |
| SLC1A5 / Glutamate–Cysteine | SLC1A5      | Moderate | 13            | 8/13 (62%) | −0.18       | GSH (reduced), Cysteine, Methionine         |